

Christopher Solanilla

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Experience

Game Programmer — CCIE

May 2025 - Present

Community Innovation and Education (CCIE) | Orlando, FL

- Redesigned and implemented a puzzle game to challenge spatial awareness using the Unity game engine.
- Built a mail server and feedback system so teachers can review student performance; deployed and hosted on AWS.

Software Engineer — Techrangers

May 2024 - Present

Center for Distributed Learning, University of Central Florida | Orlando, FL

- Developed full-stack web applications using modern JavaScript/TypeScript with React, Svelte, Vue, PHP, SQL, Docker, and AWS.
- Created a Materia widget, *Connections*, inspired by the NYT game, including authoring tools and scoring.
- Played a major role in rewriting open-source *Materia* with a Django backend (Previously Fuel PHP backend).

Software Engineer — RGS Healthcare

Dec. 2022 - Sep. 2023

RFA Medical RGS Healthcare | Miami, FL

- Engineered web scraping scripts to aggregate contact information across diverse sites, expanding the contact database.
- Designed a relational database focused on endocrinologists to enable targeted email campaigns.
- Implemented algorithms to streamline bulk email delivery to 300,000+ doctors while adhering to Google spam guidelines.

Data Engineer Intern — MD Electronics

Apr. 2021 - Sep. 2021

Ablation Technologies MD Electronics | Miami, FL

- Created and maintained a database of 600,000+ customer records using SQLite in order to do complex analysis.
- Automated bulk email delivery with Python, batching to avoid spam flags. Additionally created a Svelte web app to funnel clients.

Projects

S.T.E.P.S.

C++, OpenCV, Godot4, KiCad, Embedded Systems

- Developed a dance rhythm game blending DDR-style step mechanics with computer vision pose detection to score player style; designed and fabricated custom PCB and schematics for the dance pad and integrated system.

Body Language Detector

C++, CMake, Ninja, OpenCV, GTK3, Python

- Created a desktop app that would allow visually impaired users to detect the body language of a person on their screen for video calls. This leverages a custom tuned computer vision model.

Brawlback fork

C++, PowerPC Assembly, Linux

- Contributed to an open-source rollback netcode project for *Super Smash Bros. Brawl*; improved replay reliability on Linux, fixed Vulkan crashes, and reduced desyncs.

CodeClash.app

Mithril.js, React, MongoDB, Tailwind, Socket.io

- Built an online competitive programming platform with head-to-head problem solving with an Elo ranking system.

Chrissolanilla.com

Svelte, Tailwind, Express, Socket.io, Three.js

- Built a portfolio and playground with real-time chat and multiplayer game, evolving into a solutions platform.

ChibiChain.com

Svelte, SCSS, Firebase, SpotifyAPI

- Anime-themed link aggregation with custom backgrounds, link icons, and Spotify integration; 30+ active users.

Chess Connect 3D

Three.js, WebGL, Physics, MySQL, PHP

- Contact management app with interactive 3D chess piece visualizations and a custom JavaScript physics engine.

FIEA GameJam

Unreal Engine 5, 3D bullet-hell

- Shipped a third-person 3D bullet-hell in 48 hours; built for Windows and Linux x86.

Waypaper fork

Python, GTK4, Wayland, Hyprland

- Customized Waypaper for Wayland with enhanced compatibility and accessibility for the Hyprland compositor

Education

University of Central Florida

Expected graduation: Dec. 2025

B.S. in Computer Engineering

Skills

JavaScript, Python, C, C++, Rust, Java, C#, Lua, React, SvelteKit, Next.js, Django, Flask, Express, Tailwind, GTK, Electron, Git, GitLab, GitHub (400+ commits), Linux (Gentoo and Arch daily driver), MySQL, PostgreSQL, Docker, Mithril.js, WebGL, Three.js

Fluent in English & Spanish, proficient in Japanese